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Review

Thermodynamic reassessment of Gd–Ni system

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ABSTRACT

By means of CALPHAD (CALculation of PHAse Diagrams) approach, the phase diagram and thermodynamic data of the Gd–Ni system were critically assessed. The Gd–Ni system contains four solution phases (liquid, face-centered cubic FCC_A1, body-centered cubic BCC_A2 and hexagonal close-packed HCP_A3) modeled with the Redlich-Kister polynomials and seven intermetallic compounds Gd_3Ni, GdNi, $GdNi_2$, $GdNi_3$, Gd_2Ni_7, $GdNi_5$ and Gd_2Ni_{17}, which are all treated as stoichiometric compounds. A set of self-consistent thermodynamic parameters describing various phases in this binary system was obtained. The phase diagram and thermodynamic quantities calculated from assessed parameters agree well with experimental data.

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1. Introduction

The intermetallic compounds formed by rare earth (RE) elements and transition metals (TM) are of particular interest regarding to their potential usage as high value functional materials, such as permanent magnets [1,2] and hydrogen storage materials (reversible absorption of a large quantity of hydrogen gas at room temperature and nearly at atmospheric pressure) [3,4]. Moreover, many ternary Al–TM–RE systems form amorphous alloys with interesting mechanical properties [5] and some rare-earth/transition metal oxides are candidate materials for solid oxide fuel cells [6]. Furthermore, the rare-earth/3d transition metal intermetallic compounds were suggested as being promising candidates for room temperature magnetic refrigeration based on magneto-caloric effect (MCE) [7,8] who has attracted more attention for improved energy efficiency and environmental friendliness compared with traditional compression/expansion gas refrigeration [9,10].

On the other hand, the knowledge of thermodynamic data and phase diagrams is essential for developing and checking models of phase transformations, coupling thermodynamics with kinetics [11]. The purpose of the present work is (1) to evaluate recent experimental phase diagram and his relative thermodynamic data and (2) to provide a set of self-consistent parameters for calculation of the phase equilibria and thermodynamic properties in the Gd–Ni binary system using the CALPHAD method [12] and the Thermo-Calc software package [13].

2. Thermodynamic models

2.1. Pure elements

The stable forms of the pure elements at 298.15 K and 1 bar were chosen as the reference states of the system. For the thermodynamic functions of the pure elements in their stable and metastable states, the phase stability for the element i in ϕ status is given as:

$$\begin{aligned} {}^0G_i^{\phi}(T) &= G_i^{\phi}(T) - H_i^{SER} \\ &= a + bT + cT\,\mathrm{Ln}\,T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \end{aligned} \quad (1)$$

where $H_i^{SER}(298.15K)$ is the molar enthalpy of the so-called "Standard Element Reference" (SER), i.e., the enthalpies of the pure elements in their defined reference state at 298.15 K and 1 bar; T is the absolute temperature; $G_i^{\phi}(T)$ is the absolute molar Gibbs energy of the element i (i = Gd and Ni) with structure ϕ in a non magnetic states.

In thermodynamic study, absolute energy is not of importance, so the relative value of Gibbs energy ${}^0G_i^{\phi}(T)$ is adopted in CALPHAD approach. The Gibbs energy of the element i, ${}^0G_i^{\phi}(T)$, in its SER state is denoted by $GHSER_i$, i.e.

$$GHSER_{Gd} = {}^0G_{Gd}^{hcp}(T) = G_{Gd}^{hcp}(T) - H_{Gd}^{SER}(298.15K) \quad (2)$$

$$GHSER_{Ni} = {}^0G_{Ni}^{fcc}(T) = G_{Ni}^{fcc}(T) - H_{Ni}^{SER}(298.15K) \quad (3)$$

In the present work, the Gibb energy for the pure element Ni was taken from the SGTE compilation of Dinsdale [14], while that

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relative to the pure element Gd was taken from the work of Huasen et al. [15] who has evaluated the properties of the Gd element. We note that Dinsdale [16] has recently updated his SGTE-Pure database to incorporate the work of Huasen et al. [15].

2.2. Solution phases

The substitutional solution model was employed to describe the solution phases including Liquid, FCC_A1, BCC_A2 and HCP_A3. The molar Gibbs energy of the solution phase ϕ (ϕ = Liquid, FCC_A1, HCP_A3 and BCC_A2) can be expressed as:

$$G_m^{\phi} = x_{Gd}{}^0G_{Gd}^{\phi} + x_{Ni}{}^0G_{Ni}^{\phi} + RT(x_{Gd}Lnx_{Gd} + x_{Ni}Lnx_{Ni}) + {}^EG_m^{\phi} + {}^{mg}G_m^{\phi} \quad (4)$$

where G_m^{ϕ} is the molar Gibbs energy of a solution phase ϕ; ${}^0G_i^{\phi}$ is the molar Gibbs energy of the element i (i = Gd or Ni) with the structure ϕ in a non-magnetic state; x_i the mole fraction of component i, R gas constant, T temperature; ${}^0G_m^{\phi}$ the excess Gibbs energy, and ${}^0G_m^{\phi}$ is the magnetic contribution to the Gibbs energy. The excess Gibbs energy of phase ϕ can be expressed by the Redlich–Kister polynomials [17] as:

$${}^EG_m^{\phi} = x_{Gd}x_{Ni}\sum_j {}^jL_{Gd,Ni}^{\phi}(x_{Gd} - x_{Ni})^j \quad (5)$$

here ${}^jL_{Gd,Ni}^{\phi}$ (j = 0, 1, 2, …) is the interaction parameter between elements Gd and Ni and is formulated as temperature dependent:

$${}^jL_{Gd,Ni}^{\phi} = a_j + b_jT + c_jTLnT + d_jT^2 + e_jT^3 + f_jT^{-1} + g_jT^7 + hjT^{-9} \quad (6)$$

where a_j, b_j, c_j, d_j, e_j, f_j, g_j and h_j are model parameters to be optimized. In most cases, only the two first terms of the above equation are used.

2.3. Intermetallic compounds

All the intermetallic compounds α-Gd_3Ni, β-Gd_3Ni, GdNi, $GdNi_2$, $GdNi_3$, Gd_2Ni_7, $GdNi_5$ and Gd_2Ni_{17} were treated as stoichiometric phases in the Gd–Ni binary system because no available experimental data reports the existence of an homogeneity range for these compounds. The Gibbs energy of a Gd_ANi_B compound is given as:

$${}^0G_m^{Gd_ANi_B} = \frac{A}{A+B}{}^0G_{Gd}^{hcp} + \frac{B}{A+B}{}^0G_{Ni}^{fcc} + a + bT + {}^{mg}G_m^{Gd_ANi_B} \quad (7)$$

where ${}^0G_{Gd}^{hcp}$ and ${}^0G_{Ni}^{fcc}$ are the Gibbs energies of the respective pure elements Gd and Ni in the non-magnetic hcp and fcc structure respectively. The parameters a and b are evaluated in the present work. ${}^{mg}G_m^{Gd_ANi_B}$ is the magnetic contribution to the Gibbs energy.

As can be seen in Table 1, the Curie temperatures of the intermetallic compounds in the Gd–Ni system, available in the literature, are much below 298.15 K. For this reason, we have not taken into account the contribution of the magnetic term in our calculations.

Table 1
Curie temperatures of some intermediate phases in the Gd–Ni system.

Intermetallic compound	$GdNi_5$	GdNi	$GdNi_2$	$GdNi_3$
Curie temperature (K) [Ref.]	31 [18]	69 [19]	80 [20]	114 [21]

3. Bibliographic description of the Gd–Ni system

3.1. Equilibrium diagram

The Gd–Ni phase diagram has been investigated in 1961 by Spedding et al. [22], Novy et al. [23] and Copeland and Kato [24]. The three diagrams are in a general agreement, showing two congruent melting compounds and three eutectic reactions. Note that Gd_2Ni_{17} with a stoichiometric ratio of 2:17 reported by Novy et al. [23] was widely accepted although a ratio of 2:15 proposed by Copeland et al. [24].

Buschow and Van Der Goot [25] examined annealed samples of Gd_2Ni_7 for 2–4 weeks. They found that the main phase was hexagonal when samples were quenched from 1200 °C and rhombohedral when annealed at 700 °C. Consequently, they suggested that the hexagonal structure is the high-temperature form, and the rhombohedral type is the low-temperature form. Both of these structures always coexisted in the studied samples, and the temperature of the allotropic transition was not determined.

The Gd–Ni phase diagram was experimentally reinvestigated in 1986 by Pan et al. [26] using X-ray diffraction (XRD) and differential thermal analysis (DTA) and assessed in 1991 by Pan and Nash [27]. The following nine phases were supposed to exist: Gd_3Ni, Gd_3Ni_2, GdNi, $GdNi_2$, $GdNi_3$, Gd_2Ni_7, $GdNi_4$, $GdNi_5$ and Gd_2Ni_{17}. Among these compounds, GdNi and $GdNi_5$ melt congruently, while the others are formed by peritectic reactions. Three eutectic reactions occur: L ↔ Gd_3Ni + Gd_3Ni_2 (635 °C, ~2 at% Ni), L ↔ $GdNi_2$ + $GdNi_2$ (880 °C ~ 56 at% Ni) and L ↔ Gd_2Ni_{17} + fcc-Ni (1275 °C, ~95 at% Ni). Neither Gd in Ni nor Ni in Gd exhibits any detectable solid solubility [26].

The experimental results obtained by Pan et al. [26] and by Copeland and Kato [24] have been used for a thermodynamic assessment carried out by Xia and Jin [28] using the CALPHAD technique. However, the results obtained by these authors present large discrepancies with the experimental data, and the calculated enthalpies of formation are too less negative and seem to be inconsistent.

Using XRD at room temperature, DTA, electromotive force (EMF) measurements, Dischinger and Schaller [29] investigated this phase diagram over the entire range of composition. In comparison with the data of Pan et al. [26] there is an excellent accordance concerning the peritectic and the eutectic temperatures and a good accordance concerning the melting point of the compound $GdNi_5$. However, a discrepancy of about 300 K exists concerning the melting point of the GdNi compound.

According to Copeland and Kato [24], the Gd_3Ni_2 and $GdNi_4$ phases reported by Novy et al. [23] and Pan et al. [26] do not exist. This result was confirmed in a recent study of the ternary Gd–Ni–V system by Zhong et al. [30]. Similarly, no evidence was found to support the existence of Gd_3Ni_2 and $GdNi_4$ in the experimental study of the ternary Gd–Mn–Ni system by Yinghong et al. [31].

Coplend and Kato [24] showed some solubility of Ni in Gd, Gd in Ni, and Ni in $GdNi_5$, but no detailed information was given. Other investigators did not observe these solubilities. All intermetallic phases were suggested to be a line compound by Ref. [23] [26], and [29].

More recently, the Gd–Ni phase equilibria was critically reinvestigated by Xu et al. [32] using scanning electron microscopy with energy-dispersive X-ray spectrometry (SEM/EDS), XRD and differential scanning calorimeter (DSC) measurements realized on samples over the entire composition range. Except the confirmation of the absence of Gd_3Ni_2 and $GdNi_4$ compounds, the shape of this recent phase diagram is consistent with the results of Dischinger and Schaller [29]. However, discrepancies with the results of Pan et al. [26] was noted especially regarding the congruent melting temperature of GdNi. In the work by Xu et al. [32], the DSC curve for the Gd-20 at% Ni sample exhibits a small pic at 911 K attributed by these authors to an allotropic transformation of Gd_3Ni.

3.2. Thermodynamic data

Many researchers investigated the thermodynamic properties of Gd–Ni intermetallic compounds. The enthalpies of mixing of liquid alloys were determined at 1750 K over the entire range of composition by Nikolaenko [33]. Enthalpies of formation for several Gd–Ni intermetallic compounds have been determined by calorimetry by Colinet et al. [34], Schott and Sommer [35] and Deodhar and Ficalora [36]. Using direct synthesis calorimetry, Guo and Kleppa [37,38] measured the standard enthalpies of formation of GdNi and $GdNi_5$ compounds. Nourry et al. [40] measured experimentally the Gibbs energies of formation of some Gd–Ni intermetallic compounds. The predicted enthalpies of formation of Gd–Ni alloys from the semi-empirical model of Miedema et al. [39] are also available in the literature.

3.3. Crystallographic data

The available crystallographic data of Gd–Ni intermediate phases are compiled in Table 2.

4. Assessment procedure

The thermodynamic optimization of the model parameters of the Gibbs energy expressions is an application of the CALPHAD technique with the help of the PARROT module of the Thermo-Calc software developed by Jansson [49] and Sundman et al. [13]. Its procedure consists of the choice of thermodynamic models for the Gibbs energy of the individual phases as previously described, the analysis of the all related experimental data available, and the computer-aided nonlinear for minimizing the square sum of the errors between the experimental data and the computed values.

In the beginning of the assessment, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits. The optimization was carried out by steps. The thermodynamic parameters for the intermetallic compounds were optimized at the first stage based on the available experimental standard enthalpies of formation for the Gd–Ni intermetallic compounds and the phase boundaries information. Then experimental data related to invariant equilibria have been added to the calculation. All the parameters were finally evaluated together to give the best description of the system. We note that we have neglected the solubilities in the FCC_ A1, BCC_A2 and HCP_A3 terminal solid solutions in accord with the recent experimental results [32]. This was realized by assigning a large positive value to the ${}^{j}L^{\phi}_{Ni,Gd}$ interaction parameter of Eq (5), i.e.

$$ {}^{0}L^{HCP_A3}_{Gd,Ni} = {}^{0}L^{BCC_A2}_{Gd,Ni} = {}^{0}L^{FCC_A1}_{Gd,Ni} = 10^4 \quad (8) $$

Table 3
Thermodynamic parameters of the Gd–Ni binary system.

Phase	Thermodynamic parameters[a]
Liquid	${}^{0}L_{Gd,Ni} = -155033.136 + 61.479 \times T$
	${}^{1}L_{Gd,Ni} = 85320.912 - 36.759 \times T$
FCC_A1	${}^{0}L_{Gd,Ni} = 10^4$
HCP_A3	${}^{0}L_{Gd,Ni} = 10^4$
BCC_A2	${}^{0}L_{Gd,Ni} = 10^4$
Gd_3Ni_LT	$G_{Gd_3Ni_LT} = 0.75\,{}^{0}G^{hcp}_{Gd} + 0.25\,{}^{0}G^{fcc}_{Ni} - 16661 + 3.730 \times T$
Gd_3Ni_HT	$G_{Gd_3Ni_HT} = 0.75\,{}^{0}G^{hcp}_{Gd} + 0.25\,{}^{0}G^{fcc}_{Ni} - 16304 + 3.338 \times T$
GdNi	$G_{GdNi} = 0.50\,{}^{0}G^{hcp}_{Gd} + 0.50\,{}^{0}G^{fcc}_{Ni} - 31426 + 6.721 \times T$
$GdNi_2$	$G_{GdNi_2} = 0.33\,{}^{0}G^{hcp}_{Gd} + 0.67\,{}^{0}G^{fcc}_{Ni} - 36015 + 9.772 \times T$
$GdNi_3$	$G_{GdNi_3} = 0.25\,{}^{0}G^{hcp}_{Gd} + 0.75\,{}^{0}G^{fcc}_{Ni} - 35127 + 10.012 \times T$
Gd_2Ni_7	$G_{Gd_2Ni_7} = 0.22\,{}^{0}G^{hcp}_{Gd} + 0.78\,{}^{0}G^{fcc}_{Ni} - 34819 + 10.345 \times T$
$GdNi_5$	$G_{GdNi_5} = 0.17\,{}^{0}G^{hcp}_{Gd} + 0.83\,{}^{0}G^{fcc}_{Ni} - 29767 + 8.271 \times T$
Gd_2Ni_{17}	$G_{Gd_2Ni_{17}} = 0.11\,{}^{0}G^{hcp}_{Gd} + 0.89\,{}^{0}G^{fcc}_{Ni} - 21404 + 7.063 \times T$

[a] Gibbs energies are expressed in (J/mol.at.).

5. Results and discussion

Thermodynamic parameters for all condensed phases in the Gd–Ni binary system obtained in the present work are summarized in Table 3. In Table 4, we give invariant reactions data for this binary system. Good agreement has been realized between the calculated results and the recent experimental data reported by Xu et al. [32].

The calculated phase diagram of the Gd–Ni system is illustrated in Fig. 1. In Fig. 2 we compare our calculated phase diagram with the experimental data reported by Xu et al. [32], Pan et al. [26] and Dischinger and Schaller [29]. The phase diagram obtained in this

Table 2
Gd–Ni Crystal structures data.

Phase	Pearson symbol	Type	Symbol used in Thermo-Calc data file	Reference
α-Gd[d]	hP2	Mg	HCP_A3	[41]
β-Gd[c]	cI2	W	BCC_A2	[41]
Gd_3Ni	oP16	Fe_3C	Gd_3Ni_LT	[42]
Gd_3Ni	–	–	Gd_3Ni_HT	[32]
GdNi	oC8	CrB	GdNi	[43]
$GdNi_2$	cF24	Cu_2Mg	$GdNi_2$	[44]
$GdNi_3$	hR24	$PuNi_3$	$GdNi_3$	[45]
Gd_2Ni_7	hP36 hR54	Ce_2Ni_7[a] Gd_2Co_7[b]	Gd_2Ni_7	[25]
$GdNi_5$	hP6	$CaCu_5$	$GdNi_5$	[46]
Gd_2Ni_{17}	hP38	Th_2Ni_{17}	Gd_2Ni_{17}	[47]
Ni	cF4	Cu	FCC_A1	[48]

[a] High-temperature form.
[b] Low-temperature form.
[c] From 1508 K–1586 K.
[d] Below 1508 K.

Table 4
Calculated and experimental invariant equilibria in the Gd–Ni system.

Reaction	Type	Temperature (K)		Composition of the liquid phase (at. % Ni)	
		This work	Literature data [Ref.]	This work	Literature data [Ref.]
Liquid + α-Gd ↔ Gd_3Ni_HT	Peritectic	1039	1039 [32] 986 [29] 1008 [26] 1006 [28]	32	30 [32] 28 [28]
Gd_3Ni_LT ↔ Gd_3Ni_HT	Allotropic transition	911	911 [32]	–	–
Liquid ↔ Gd_3Ni_HT + GdNi	Eutectic	977	977 [32]	36	33 [32]
Liquid ↔ GdNi	Congruent	1238	1238 [32] 1253 [29] 1553 [26]	–	–
Liquid ↔ GdNi + $GdNi_2$	Eutectic	1172	1172 [32] 1158 [29] 1153 [26] 1232 [28]	56	56 [32] 56 [26] 57 [28]
Liquid + $GdNi_3$ ↔ $GdNi_2$	Peritectic	1358	1358 [32] 1268 [29] 1283 [26] 1280 [28]	64	61 [32] 58 [26] 57 [28]
Liquid + Gd_2Ni_7 ↔ $GdNi_3$	Peritectic	1481	1480 [32] 1383 [29] 1383 [26] 1372 [28]	69	66 [32] 61 [26] 60 [28]
Liquid + $GdNi_5$ ↔ Gd_2Ni_7	Peritectic	1556	1557 [32] 1473 [29] 1473 [26] 1474 [28]	73	71 [32] 65 [26] 64 [28]
Liquid ↔ $GdNi_5$	Congruent	1705	1706 [29] 1753 [26] 1760 [28]	–	–
Liquid + $GdNi_5$ ↔ Gd_2Ni_{17}	Peritectic	1589	1589 [32] 1573 [29] 1558 [26] 1553 [28]	91	90 [32] 94 [26] 95 [28]
Liquid ↔ Gd_2Ni_{17} + Ni	Eutectic	1571	1571 [32] 1563 [29] 1548 [26] 1521 [28]	93	94 [32] 95 [26] 96 [28]

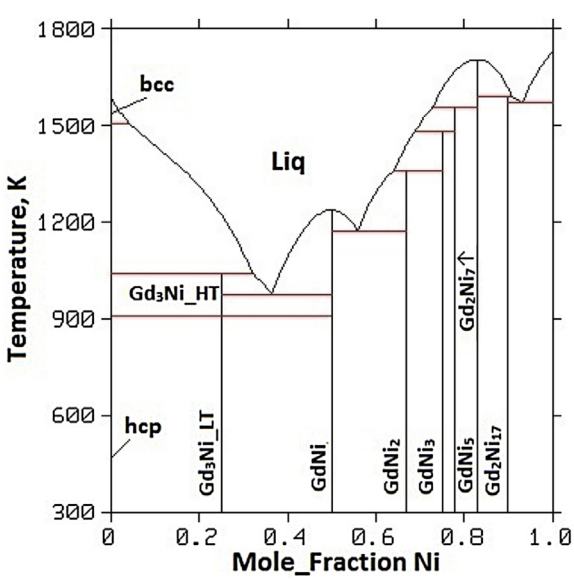

Fig. 1. The Gd–Ni phase diagram calculated in this work.

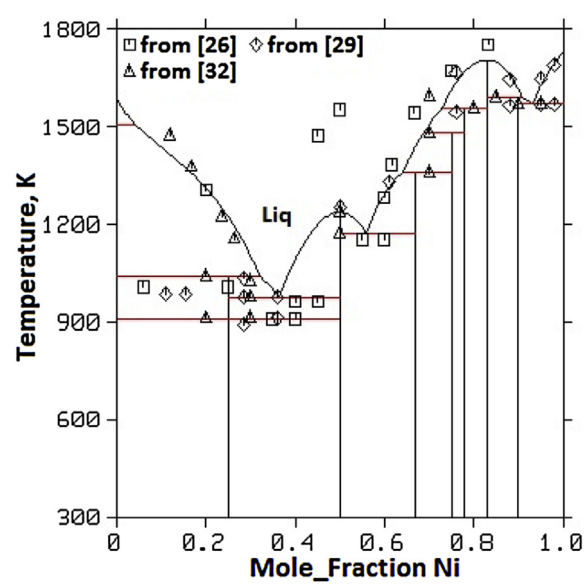

Fig. 2. Calculated Gd–Ni phase diagram compared with available experimental data.

work is in good agreement with that established recently by Xu et al. [32]. The incompatible asymmetry of the liquidus phase [50,51] on the left and right sides of the $GdNi_5$ compound was corrected in the course of this study (Fig. 2). Indeed, according to Okamoto and Massalski's studies [50,51], judgment about asymmetry of liquidus curves can be made comparing the ratio of the width of the two phase fields on each side of a compound at the same temperature. If the width on one side of the compound is either one half or double that on the other side, the asymmetry may be considered clearly unusual.

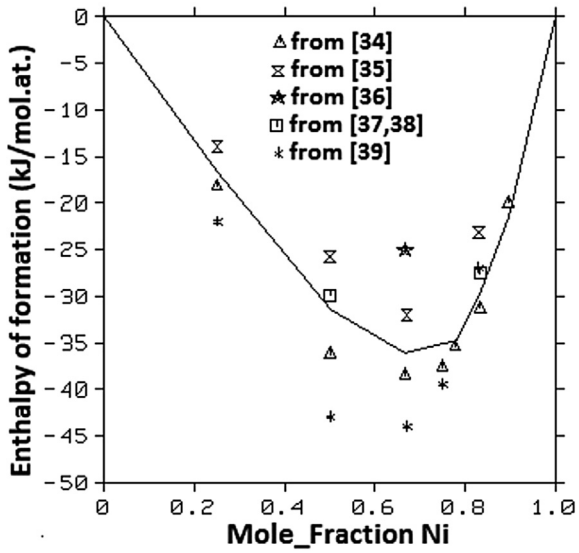

Fig. 3. Calculated enthalpies of formation of the Gd–Ni intermetallic compounds with available literature data.

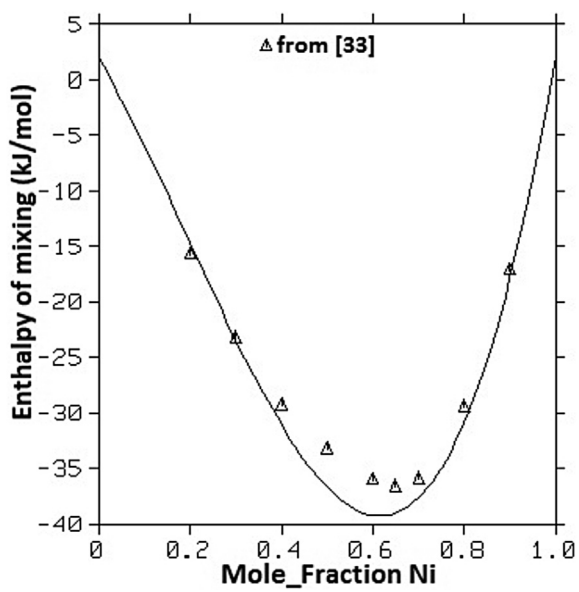

Fig. 4. Calculated enthalpy of mixing of the liquid phase at 1750 K. Experimental data are the calorimetric results from Ref. [33].

As mentioned previously, the enthalpies of formation of the Gd–Ni intermetallic compounds are measured by several authors. In general, direct calorimetric measurements are considered to be the most reliable, so the values obtained by Guo and Kleppa [37,38] were used in the present study with high weights. Fig. 3 shows the standard enthalpies of formation of Gd–Ni intermediate phases together with the predicted values by the Miedema model [39] and with the experimental data determined by Colinet et al. [34], Schott and Sommer [35], Deodhar and Ficalora [36] and Guo and Kleppa [37,38]. A reasonable agreement is noted between the calculated results and the experimental values determined by calorimetry. The enthalpies of formation for many compounds predicted by Miedema model [39] are usually more exothermic compared to the calorimetric data and they may be considered using precaution. The enthalpy of formation determined by Deodhar and Ficalora [36] for $GdNi_2$ using calorimetry [36] is less negative than our calculated one. This value is also less negative compared to the other

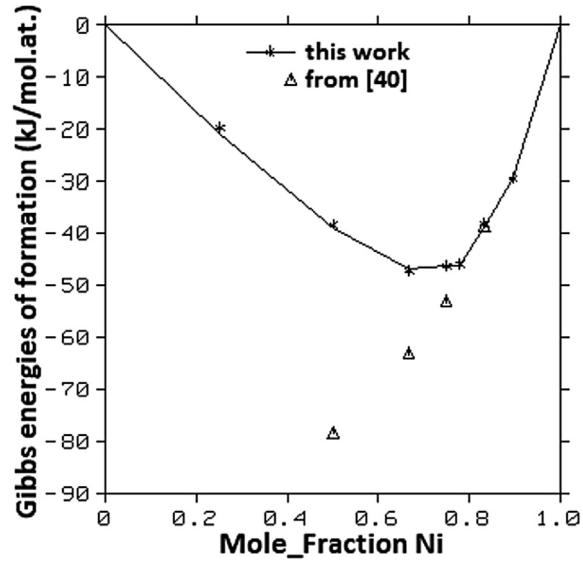

Fig. 5. Gibbs energies of formation of Gd–Ni intermetallic compounds at 1113 K.

experimental values obtained by Colinet et al. [34] and Schott and Sommer [35]. The discrepancy between the experimental results is certainly due to the relative purity of the starting materials [52,53] and to the fact that the rare earth elements are largely reactive with crucibles and containers, particularly at high temperatures [54].

We note that the enthalpies of formation of Gd_3Ni and $GdNi_2$ calculated by Xia and Jin [28] are too less negatives and differ from the experimental values of Colinet et al. [34] by 13.5 kJ/mol and 23.7 kJ/mol respectively.

Fig. 4 presents the calculated enthalpy of mixing of liquid at 1750 K in the Gd–Ni system with the experimental data determined by Nicolaenko [33]. The calculated values are satisfactorily consistent with the measured data in Ref. [33].

Gibbs energies of formation of intermetallic compounds at 1113 K were further calculated and compared with data measured by Nourry et al. [40] in Fig. 5. Good agreement was realized between the calculated values and experimental results for $GdNi_5$ and $GdNi_3$ intermetallic compounds. However, Gibbs energies of GdNi and $GdNi_2$ at 1113 K are −78.5 and −66.3 (kJ/mol. at.) respectively while the assessed values are −38.3 and −47.2 (kJ/mol. at.). The disagreement is noticeable.

On the other hand, if we considered both the enthalpies of formation measured by calorimetry and the Gibbs energies of formation determined by Nourry et al. [40] for the GdNi and $GdNi_2$ intermetallic compounds, the relative entropies of formation must become largely negative, which is not reasonable from thermodynamic viewpoint. Thus, we the think that the Gibbs energy of formation of GdNi and $GdNi_2$ determined by Nourry et al. [40] are not reliable.

6. Conclusion

The experimental phase equilibria and the thermodynamic data relative to the Gd–Ni system available in the literature have been critically evaluated. Within the scheme of the CALPHAD technique, the thermodynamic models for all the solution phases and intermediate compounds were selected and the Gibbs energy functions were optimized. A set of consistent thermodynamic parameters was obtained for the Gd–Ni system. The calculated phase equilibria agree well with the recent experimental one established by Xu et al. [32]. The thermodynamic data, such enthalpies of mixing of liquid, standard enthalpies and Gibbs free energies of formation of the Gd–Ni intermetallic compounds, were confronted to the available values from the literature.

Appendix

The Gibbs free energies of the pure elements (solid and liquid) in the stable and metastable states, taken from SGTE database [16]. The data are given in J/mol of atoms.

Nickel

GHSERNi

$= -5179.159 + 117.854T - 22.096T\mathrm{Ln}(T) - 0.0048407T^2$ (298.15 <*T*< 1728.00)

$= -27840.655 + 279.135T - 43.1T\mathrm{Ln}(T) + 1.12754.10^{31}T^{-9}$ (1728.00 <*T*< 3000.00)

GLIQNi

$= +16414.686 - 9.397T + \mathrm{GHSERNi} - 3.82318.10^{-21}T^7$ (298.15 <*T*< 1728.00)

$= -9549.775 + 268.598T - 43.1T\mathrm{Ln}(T)$ (1728.00 <T< 3000.00)

Gadolinium

GHSERGd

$= -6834.5855 + 97.13101T - 24.7214131T\mathrm{Ln}(T) - 0.00285240521T^2 - 3.14674076.10^{-7}T^3 - 8665.73348T^{-1}$ (200.00 <T< 1000.00)

$= -6483.25362 + 95.6919924T - 24.6598297T\mathrm{Ln}(T) - 0.00185225011T^2 - 6.61211607.10^{-7}T^3$ (1000.00 <T< 1508.15)

$= -123124.992 + 699.125537T - 101.800197T\mathrm{Ln}(T) + 0.0150644246T^2 - 6.39165948.10^{-7}T^3 + 29356890.3T^{-1}$ (1508.15 <*T*< 3600.00)

GLIQGd

$= +13060.0262 - 8.32179974T + \mathrm{GHSERGd}$ (200.00 <*T*< 1000.00)

$= +146262.037 - 1208.70685T + 159.352082T\mathrm{Ln}(T) - 0.108247135T^2 + 1.06945505.10^{-5}T^3 - 19678357T^{-1}$ (1000 <*T*< 1508.15)

$= -5397.314 + 192.336215T - 38.5075T\mathrm{Ln}(T)$ (1508.15 <*T*< 3600.00)

GBCCGd

$= +3233.80866 - 2.11184585T + 1.00000001.10^{-7}T\mathrm{Ln}(T) + \mathrm{GHSERGd}$ (200.00 < T< 1000.00)

$= +152792.743 - 1349.58873T + 180.097094T\mathrm{Ln}(T) - 0.119550229T^2 + 1.17915728.10^{-5}T^3 - 22038836T^{-1}$ (1000.00 <*T*< 1508.15)

$= -15783.7618 + 202.222057T - 38.960425T\mathrm{Ln}(T)$ (1508.15 <*T*< 1586.15)

$= -19850.5562 + 224.817909T - 41.904333T\mathrm{Ln}(T) + 8.58222759.10^{-4}T^2 - 3.77570269.10^{-8}T^3 + 995428.573T^{-1}$ (1586.15 <*T*< 3600.00)

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